Non-linear Network Analysis of Extreme Precipitation Propagation in Southern India: The Impact of ENSO Phases

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Abstract. Understanding the spatiotemporal dynamics of Extreme Precipitation Events (EPEs) during the Indian Summer Monsoon (JJAS) is critical for regional climate risk assessment. This study investigates the characteristics of EPEs over Southern India using complex climate network analysis. To ensure topography-aware detection of extremes, the Multiple Threshold Anderson-Darling (MT-AD) method is applied, avoiding the overestimation inherent in empirical percentile methods. Based on the robustly detected EPEs, Event Synchronization (ES) is utilized to construct the climate network. The topological analysis reveals a distinct source-to-sink architecture: the Western Ghats acts as a localized orographic source with low connectivity, whereas the North Interior Karnataka (NIK) region serves as a prominent synoptic propagation hub with high network divergence and degree. Furthermore, the network topology is highly modulated by the El Niño-Southern Oscillation (ENSO). During the El Niño phase, the precipitation network becomes fragmented and weakened. Conversely, the La Niña phase facilitates highly organized and robust propagation pathways, demonstrating a total network degree approximately 1.5 times higher than that of El Niño. These findings provide a statistical and topological basis for advancing localized early warning systems and understanding moisture propagation dynamics in Southern India.

1 Introduction

1.1 Motivation

The Indian Summer Monsoon Circulation is the dominant climatic factor determining precipitation patterns over the Indian subcontinent. The southwest monsoon, occurring between June and September (JJAS), serves as the primary rainy season. Extreme Precipitation Events (EPEs) during this period are critical subjects for socio-economic climate risk assessments, including flooding and agricultural productivity. Consequently, a precise understanding of the occurrence mechanisms and propagation pathways of EPEs is essential for establishing regional climate adaptation strategies.

1.2 State of the Art

Accurate identification of EPEs is a fundamental prerequisite for understanding their spatiotemporal dynamics. Recent research has moved toward robust statistical models, shifting away from empirical percentile-based approaches. For the Indian subcontinent, Sree Anusha and Maheswaran (2025) demonstrated that the Peak Over Threshold (POT) approach coupled with the

Generalized Pareto Distribution (GPD) provides a topography-aware framework for threshold selection that is more effective than traditional methods.

To capture the non-linear connectivity and propagation of these identified EPEs, Event Synchronization (ES) and complex network theory have been increasingly employed. Malik et al. (2010) utilized ES to reveal the coherent spatial structures and time-delay directionalities of the South Asian monsoon. Subsequently, Boers et al. (2016) integrated ES with complex networks and statistical null models to analyze regional connectivity (RC) by classifying EPEs based on intensity, duration, and spatial extent. Recently, Li et al. (2024) applied the concept of network divergence (ΔS) within global climate networks to explicitly map the source-to-sink propagation pathways of EPEs.

Furthermore, the topological characteristics of these networks are heavily influenced by large-scale climate forcings. Sahany et al. (2018) established the influence of ENSO and Indian Ocean sea surface temperatures on the spatiotemporal variability and seasonality of Indian rainfall. Recent studies have actively integrated complex network topology with meteorological dynamics. Gupta et al. (2022) identified the synchronization pathways between the Indian and East Asian summer monsoons by considering network metrics alongside atmospheric circulation anomalies and intra-seasonal oscillations (ISO). Additionally, thermodynamic analyses and Lagrangian moisture tracking are utilized to physically validate statistical network linkages, as demonstrated by Ganapathiraju et al. (2026) in evaluating moisture influx dynamics during specific EPE case studies.

1.3 Outline

Drawing upon these methodological advancements, this study investigates the precipitation network of Southern India using non-linear and complex network theories. The analysis begins with robust EPE threshold detection, followed by an examination of overall network propagation patterns. Structural variations in the network across different ENSO phases are then analyzed, and a specific case study is conducted to evaluate whether observed rainfall patterns are consistent with the derived network characteristics.

2 Data and Methodology

2.1 Data

The daily rainfall dataset for the Indian peninsular region spans from 1951 to 2023 and comprises 581 grid points with a spatial resolution of $0.25^\circ \times 0.25^\circ$. Measured in mm/day, these observations are filtered to isolate the summer monsoon season (June to September). For the EPE detection, the provided threshold values are utilized. These thresholds were derived following the Multiple Threshold Anderson–Darling (MT-AD) methodology established by Sree Anusha and Maheswaran (2025) to ensure spatial consistency across the study domain. Additionally, monthly ENSO index data for the Niño 3.4 region (1951–2023) are incorporated based on the criteria described by Sahany et al. (2018).

2.2 Methodology

The analytical framework is structured into a sequential flow to investigate extreme precipitation propagation. Threshold selection methods are evaluated to extract independent Extreme Precipitation Events (EPEs). Non-linear spatiotemporal connectivity between these events is measured using Event Synchronization (ES), parameterized by dynamic delays (τ) and a maximum delay constraint (τ_{max}). A functional climate network is then constructed through statistical filtering using a null model. Network metrics (degree and divergence) are compared across different ENSO phases to identify structural variations. A propagation score is applied to identify significant events for validation through standardized rainfall anomaly mapping.

2.2.1 Threshold approach to detect extreme precipitations events

To identify the most suitable threshold for extreme precipitation events, several methods including the Anderson-Darling (AD) and Multiple Threshold Anderson-Darling (MT-AD) and the 95th percentile of wet days are evaluated. A lower limit of 1 mm per day is applied across all methods to exclude non-significant rainfall. While the implementation of AD and MT-AD follows Sree Anusha and Maheswaran (2025), the MT-AD procedure is specifically modified to select thresholds based solely on the minimum Mean Absolute Deviation. Following the selection of the MT-AD method, time indices for days exceeding the grid wise thresholds are extracted for 581 grid points. Only the first occurrence of threshold exceedance is considered to prevent the counting of multiple peaks from the same rainfall event. This approach ensures that occurrences within a 7 day window are not treated as independent. Binary event time series are subsequently constructed through this identification process.

2.2.2 Event Synchronization

Event Synchronization (ES) is applied as a non-linear technique to quantify the degree of synchronization and time delay patterns between event time series. This method allows for the measurement of non-linear edge weights across the study area. The mathematical formulation follows Li et al. (2024) utilizing a dynamical delay τ and a maximum delay constraint τ_{max}. Although τ_{max} values of 5 and 7 days are evaluated, a constant of 7 days is selected because no significant variations in the results are observed. For a specific event μ at grid i the local time interval is defined as

$$\tau_i^{\mu} = \min(t_i^{\mu} - t_i^{\mu-1}, t_i^{\mu+1} - t_i^{\mu})$$

The dynamical delay $\tau_{i,j}^{\mu,\nu}$ for a pair of events is subsequently determined as

$$\tau_{i,j}^{\mu,\nu} = \frac{1}{2}\min(\tau_i^{\mu}, \tau_j^{\nu})$$

Synchronization between grid points i and j is evaluated through the indicator $S_{ij}^{\mu,\nu}$ as follows:

$$S_{ij}^{\mu,\nu} = \begin{cases} 1 & \text{if } 0 < t_j^{\nu} - t_i^{\mu} \leq \min(\tau_{i,j}^{\mu,\nu}, \tau_{max}) \\ 0.5 & \text{if } t_j^{\nu} = t_i^{\mu} \\ 0 & \text{otherwise} \end{cases}$$

The total directional event count $c(j|i)$ where occurrences at grid j follow those at grid i is calculated by

$$c(j|i) = \sum_{\mu=1}^{s_i} \sum_{\nu=1}^{s_j} S_{ij}^{\mu,\nu}$$

Normalization of these counts is performed to derive the synchronization strength Q_{ij} and the delay directionality q_{ij} through the following expressions

$$Q_{ij} = \frac{c(j|i) + c(i|j)}{\sqrt{s_i s_j}}$$

$$q_{ij} = \frac{c(j|i) - c(i|j)}{\sqrt{s_i s_j}}$$

In contrast to the approach by Boers et al. (2016) where simultaneous events are excluded, a weight of 0.5 is distributed to both directions when $t_i^\mu = t_j^\nu$ to preserve statistical information. Before the construction of the adjacency matrix, the internal spatial structure is analyzed following Malik et al. (2010) to identify coherent zones and propagation vectors.

2.2.3 Network Construction

A null model is employed to perform statistical significance testing on the event synchronization metrics to filter spurious occurrences. The significance level for this test is established at the 95th percentile. To construct the adjacency matrix A_{ij}, the synchronization values are binarized where significant edges are assigned a value of 1 and non-significant edges are assigned 0. Sensitivity parameters are maintained consistently with the synchronization step, including a maximum delay constraint τ_{max} of 7 days. While Li et al. (2024) and Boers et al. (2016) utilized 1000 surrogate event series, the number of surrogates is restricted to 100 in this analysis due to computational constraints.

Topological properties are quantified using the following metrics. The out-degree k_i^{out} and in-degree k_i^{in} represent the number of outgoing and incoming edges for node i, respectively. These are calculated through the summation of the adjacency matrix elements as:

$$k_i^{out} = \sum_{j=1}^{N} A_{ij}, \quad k_i^{in} = \sum_{j=1}^{N} A_{ji}.$$

The total degree k_i^{tot}, which indicates the overall connectivity hubness of a grid point, is defined by the sum of these directional components:

$$k_i^{tot} = k_i^{out} + k_i^{in}.$$

The network divergence ΔS is subsequently determined by the difference between the out-degree and in-degree:

$$\Delta S = k_i^{out} - k_i^{in}.$$

This metric is utilized to characterize each grid point as either a source or a sink within the propagation pathways.

2.2.4 ENSO Modulation Analysis

To investigate the modulatory effects of large-scale climate oscillations, a comparative analysis is performed based on the El Niño-Southern Oscillation (ENSO) phases. The classification of these phases is determined by the mean Oceanic Niño Index (ONI) during the JJAS season from 1951 to 2023. Years are categorized as El Niño if the mean JJAS index is ≥ 0.5, and as La Niña if the mean index is ≤ -0.5. Independent climate networks are constructed for each phase by subsetting the event time series to include only the days corresponding to the respective ENSO years. For each sub-network, separate null model simulations and network construction procedures are executed to ensure phase-specific statistical significance. The resulting topological properties, including the degree k_i and network divergence ΔS, are compared between the two phases to identify structural variations in propagation pathways.

2.2.5 Case Study using Propagation Score

To validate the physical plausibility of the source-to-sink pathways derived from the network analysis, a propagation score is conceptualized based on the synchronicity identification framework of Gupta et al. (2022). The top 50 source grids and the bottom 50 sink grids are identified based on the network divergence ΔS. For each day t from 1951 to 2023, the score $Score(t)$ is calculated as the product of the number of extreme precipitation events at the source grids and the number of unique sink grids that experienced events within the subsequent τ_{max} of 7 days. This calculation is expressed as follows:

$$Score(t) = N_{source}(t) \times N_{sink}^{unique}(t, t + \tau_{max}).$$

Five events with the highest scores are extracted from the total study period. The most recent case among these events is selected for detailed analysis. The trajectory of the precipitation system is subsequently verified through standardized daily rainfall anomaly maps from Lag 0 to Lag 5.

3 Results and Discussions

3.1 Threshold Detection Method

To evaluate the validity of the threshold-based approach for detecting extreme precipitation events (EPEs), the distributional characteristics of several threshold selection methods, including Anderson-Darling (AD), Multiple Threshold Anderson-Darling (MT-AD), and the 95th percentile, are compared. The spatial distributions and the corresponding grid count and density histograms are presented in **Figure 1** and **Figure 2**.

The 95th percentile method exhibits a distribution concentrated within the 35–40 mm/day range. In contrast, the Generalized Pareto Distribution (GPD) based approaches, including the AD and MT-AD methods, commonly peak around 20 mm/day and demonstrate a right-skewed distribution with a heavy tail. As discussed in Sree Anusha and Maheswaran (2025), these results indicate that the Percentile method tends to overestimate the actual threshold parameters of the population.

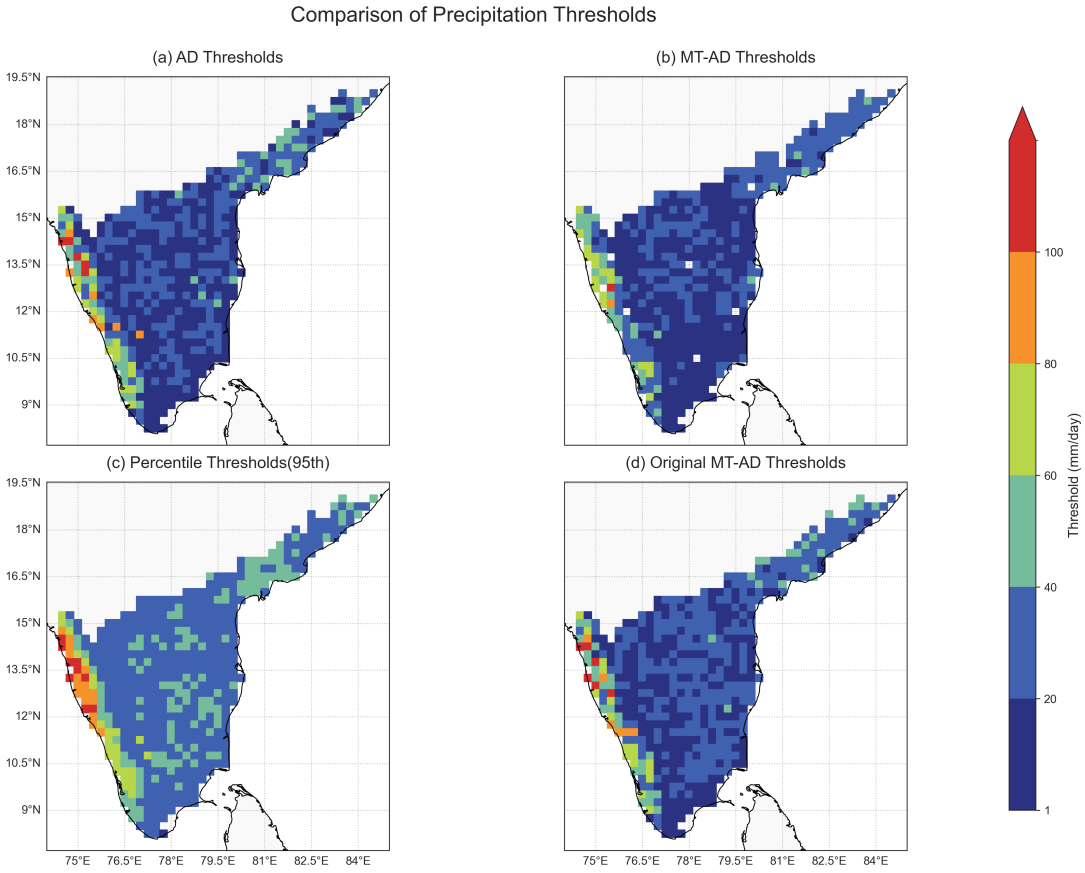


Figure 1. Spatial distribution of precipitation thresholds identified by (a) AD, (b) MT-AD, (c) 95th percentile, and (d) Original MT-AD threshold values.

The AD method (a) successfully calculates thresholds for 100% of the 578 grids without missing data, thereby ensuring spatial continuity. However, the relatively large spread in its distribution suggests a potential for overestimation or underestimation. As shown in **Figure 2**, (b) the constructed MT-AD thresholds shows a distribution highly similar to the Original MT-AD thresholds (d), indicating a successful reproduction of the methodology.

As shown in **Figure 1**, the percentile method displays a smooth and gradual variation in thresholds in the Western Ghats region. Conversely, the statistical techniques (AD and MT-AD) are found to capture thresholds with higher sensitivity at the individual grid level. Furthermore, the wider range of threshold distributions observed in the statistical methods compared to the percentile method suggests a superior capability in capturing diverse thresholds according to regional climate variability.

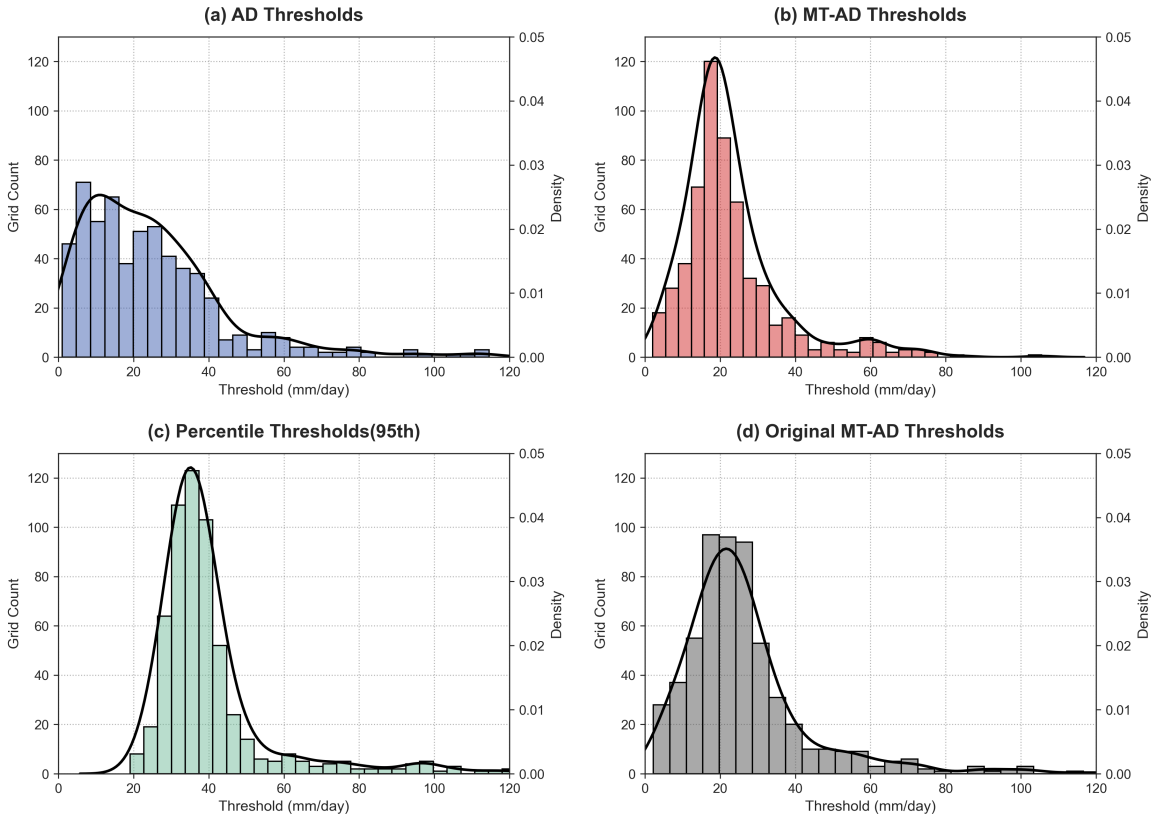


Figure 2. Histograms of grid counts and density for (a) AD, (b) MT-AD, (c) 95th percentile, and (d) Original MT-AD thresholds.

However, a failure in statistical convergence is observed in 2.42% of the total grids (14 grids) when utilizing the MT-AD approach, resulting in missing data. Given that subsequent climate network analysis requires uninterrupted spatial continuity, the original MT-AD thresholds (d) are utilized for the construction of the final binary event series.

3.2 Precipitation Network Patterns

3.2.1 Spatial Clustering and Propagation Vectors

The spatial synchronization and directionality of extreme precipitation events are analyzed using the synchronization strength matrix Q and the delay directionality matrix q. The resulting clustering and vector fields are illustrated in **Figure 3**

Clustering based on the Q matrix reveals the degree of spatial synchronization among EPEs across the study area. Six distinct coherent zones are identified through this process. Specifically, the Western Ghats region (Zones 3 and 4) is separated into

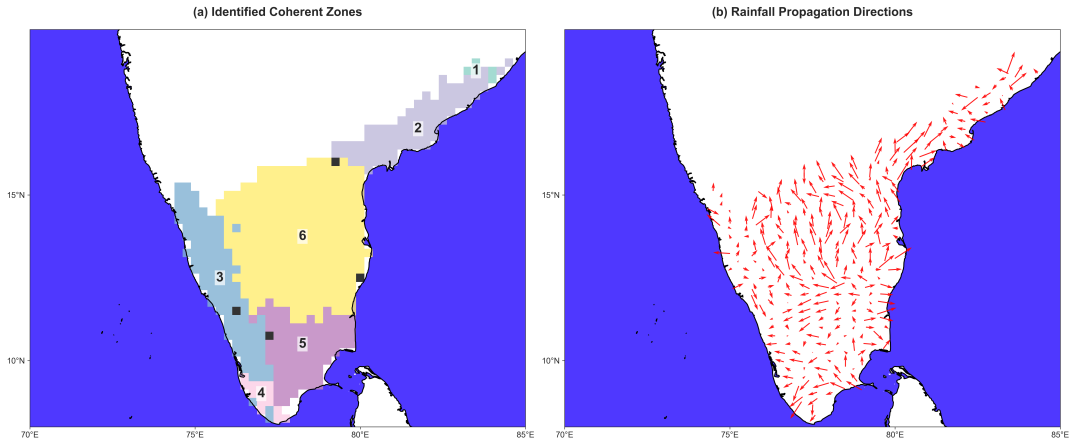


Figure 3. Spatial clustering and propagation vector fields based on the Q and q matrices.

independent clusters. This spatial organization is attributed to the humid southwesterly winds from the Arabian Sea during the JJAS period, which encounter the topographic barrier of the Western Ghats. The resulting orographic lift induces highly synchronized precipitation events along the narrow western coastal strip, which is characterized by the highest rainfall intensities in the subcontinent. As the air masses cross the mountain range, moisture is depleted and descending motion occurs over the lee side. Consequently, inland regions such as the Deccan Plateau exhibit distinct precipitation patterns and lower synchronization compared to the windward coastal regions.

The propagation vectors derived from the q matrix represent the mean trajectory and temporal precedence of the rainfall systems. In Zones 1, 2, and 6, the vectors align with the primary southwesterly monsoon flow, progressing from the southwest toward the northeast. In contrast, the vectors in the Western Ghats (Zones 3 and 4) indicate that rainfall systems do not directly penetrate the inland regions. Instead, the systems are impeded by the mountain wall, leading to stagnation or propagation along the coastline. As noted by Malik et al. (2010), the Indian summer monsoon is characterized by two major moisture sources: the Arabian Sea to the west and the Bay of Bengal to the east. In Zone 5, encompassing the central and eastern regions, the precipitation is modulated by low-pressure systems developing over the Bay of Bengal. These systems move toward the land, resulting in a pronounced vector directionality from the Bay of Bengal toward the interior of the subcontinent.

3.2.2 Network Degree and Divergence

The spatial distributions of the total network degree k_i and the network divergence ΔS are presented in **Figure 4**. These metrics characterize the topological structure and the propagation characteristics of the extreme precipitation network in Southern India.

The total network degree k_i is utilized to identify the propagation hubs within the network. A high total degree indicates that the occurrence of an extreme precipitation event (EPE) at a specific location is spatially synchronized with other regions across a broad extent. The results demonstrate that the Western Ghats and Coastal Andhra Pradesh (CAP) exhibit relatively

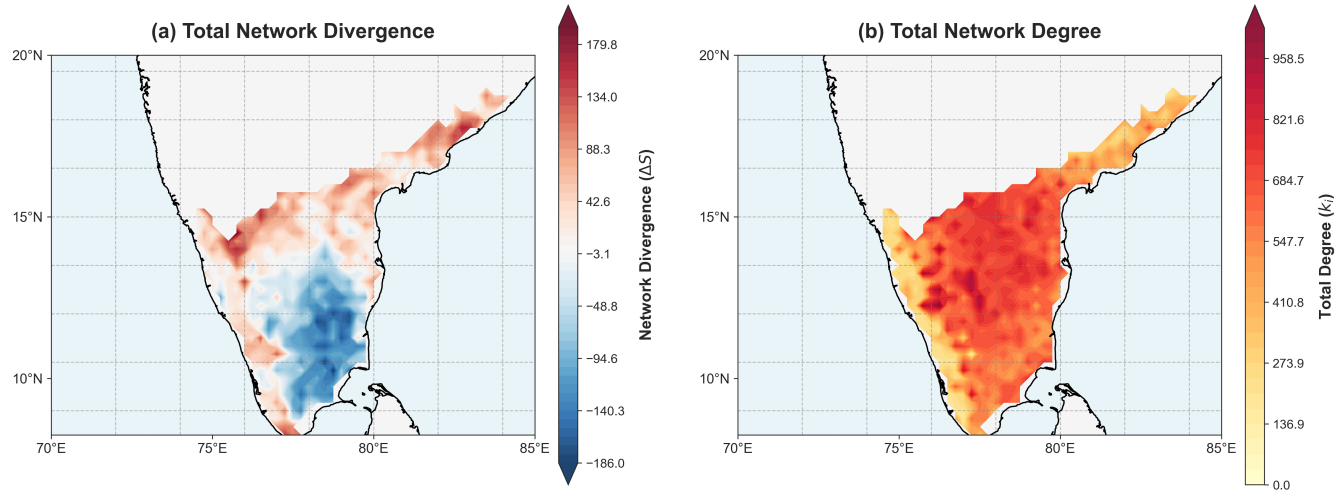


Figure 4. Network characteristics: (a) Network divergence (ΔS) and (b) Total network degree (k_i).

low degree values. In contrast, the inland regions located to the east of the Western Ghats and the entire eastern portion of the study area are characterized by high degree values.

The network divergence ΔS identifies the moisture sources and sinks, indicating the origin (source, positive) and termination (sink, negative) of extreme precipitation signals. Source regions are identified in the Western Ghats, CAP, and North Interior Karnataka (NIK). The southeastern region, specifically Tamil Nadu, is identified as a major sink.

The combined analysis of divergence and degree allows for a more detailed interpretation of these sources. The Western Ghats is characterized as a localized orographic source. While it exhibits positive divergence, the total degree remains low. This suggests that moisture originating from the Arabian Sea is intensely concentrated through orographic rainfall and subsequently dissipates due to the rain-shadow effect. Consequently, topographical isolation prevents the propagation of EPEs further inland.

Conversely, the NIK region is identified as a synoptic propagation source, displaying both high total degree and positive divergence. This indicates that the region serves as a primary corridor for synoptic systems, such as monsoon depressions from the Bay of Bengal or trans-continental systems, to propagate precipitation signals to the inland sink regions. Although CAP is the landfall point for monsoon depressions and exhibits a strong source signal, the network degree remains low because the trajectories of these systems are typically directed toward the northwest.

3.3 ENSO Modulation of the Network

The structural variations in the monsoonal precipitation network of Southern India are analyzed based on the ENSO phases. The results indicate significant topological differences between the El Niño and La Niña phases, as illustrated in the network divergence comparison (**Figure 5**).

A collapse of connectivity is observed during the El Niño phase. The total degree in the El Niño network is identified as $82,483$, which is approximately 1.5 times lower than that of the La Niña network ($119,241$). The average degree also decreases

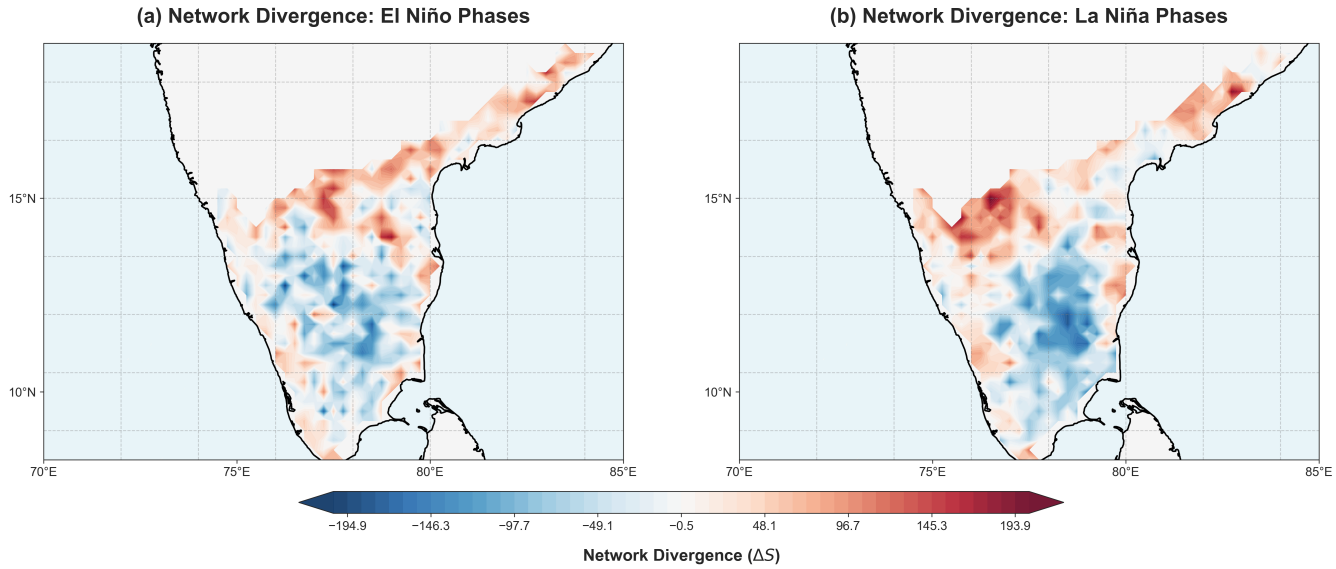


Figure 5. Comparison of network divergence (ΔS) between (a) El Niño and (b) La Niña phases.

to 285.4 in the El Niño phase compared to 412.6 in the La Niña phase. These findings suggest that the extreme precipitation propagation system in Southern India becomes fragmented and weakened during El Niño events. A comprehensive comparison of the topological properties across the different phases is summarized in Table 1.

In contrast, an activation of propagation hubs is identified during the La Niña phase. The maximum total degree reaches 828, demonstrating that specific regions perform as potent hubs within the propagation network.

The patterns of spatial divergence (ΔS) also exhibit distinct variations. During the El Niño phase, the overall intensity of divergence is weakened, leading to an ambiguous distinction between the sources and sinks of propagation (**Figure 5**a). Conversely, a pronounced structural propagation pathway is identified during the La Niña phase (**Figure 5**b). Strong positive divergence (source) is observed in the North Interior Karnataka (NIK) region, while strong negative divergence (sink) is concentrated in the southeastern inland regions, forming a clearly defined propagation route.

Table 1. Comparison of network metrics across all years and different ENSO phases.

Metric	Original (All Years)	La Niña	El Niño
Total Edges	164,380	119,241	82,483
Max Total Degree	1,027	828	583
Average Degree	568.79	412.60	285.41
Max Divergence (ΔS)	195	202	179
Min Divergence (ΔS)	−186	−181	−203

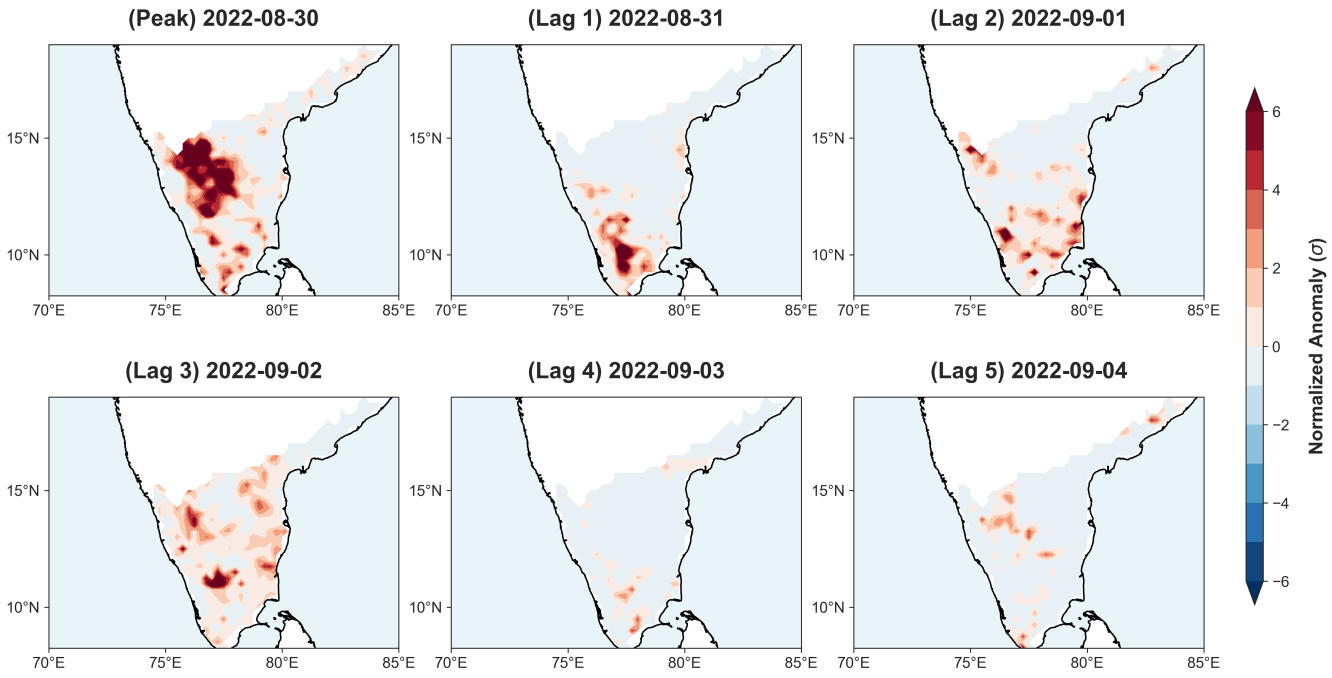


Figure 6. Spatiotemporal propagation of standardized precipitation anomalies starting from 30 August 2022 (Lag 0 to Lag 5).

3.4 Case Study: The 2022 La Niña

The top five events are identified based on the propagation score $Score(t)$. Although the highest score of 570 was recorded on 24 September 2010, the event on 30 August 2022 (Rank 4, $Score = 416$) is selected for a detailed analysis to investigate the most recent propagation dynamics. This case study is illustrated in **Figure 6**.

At Peak (30 August), extreme precipitation anomalies exceeding 6σ are observed in the North Interior Karnataka (NIK) region, which was previously identified as a source area in the network analysis. This peak corresponds to the simultaneous activation of 16 out of the 50 source grids.

From Lag 1 to Lag 5 (31 August to 4 September), the precipitation anomaly patterns demonstrate a southeastward movement and spatial expansion. Specifically, the center of the anomaly system shifts southward at Lag 1, and the regions of positive anomalies subsequently expand across the southeastern inland region, defined as the sink area, at Lag 2 and Lag 3.

The Nina index for September 2022 indicates a strong La Niña phase (-0.97). The organized propagation pattern observed during this period is consistent with the characteristics derived from the ENSO modulation analysis, namely the high connectivity and the distinct source-sink structure associated with the La Niña phase.

4 Limitations and Future Research

The findings of this study are subject to several limitations that suggest directions for subsequent research. First, the number of random shuffles for the generation of surrogate data was restricted to 100 times due to computational resource constraints. Given that established global and continental-scale network studies, such as Boers et al. (2016) and Gupta et al. (2022), typically employ $1,000$ to $2,000$ shuffles to estimate the 95% significance threshold, the statistical robustness should be strengthened in future investigations by substantially increasing the shuffling iterations.

Second, a potential inconsistency exists between the selection of the maximum time lag (τ_{max}) and the analytical spatial scale. While τ_{max} was set to 7 days based on universal synoptic meteorological standards, the spatial extent of this study is confined to Southern India ($0°$-$20°$N). The case study revealed that EPEs can traverse from source to sink regions within only 1-2 days. Consequently, maintaining a τ_{max} of 7 days may lead to an overestimation of connectivity by accounting for independent precipitation events as synchronized occurrences. Although no significant visual differences in network divergence and degree maps were observed when τ_{max} was reduced to 5 days, a more rigorous sensitivity analysis regarding smaller τ_{max} values is required.

Finally, this study analyzed the average monsoon network topology by integrating the entire period from 1951 to 2023. However, recent literature indicates that the Indian summer monsoon is undergoing changes in seasonality, frequency, and intensity of extreme precipitation due to anthropogenic climate change and rising sea surface temperatures. Therefore, future studies should analyze the study period by decade to track how the network metrics of Southern India are altered by climate change.

5 Conclusions

This study investigated the propagation characteristics of extreme precipitation events (EPEs) in Southern India during the Indian summer monsoon (JJAS) through climate network analysis. The following conclusions are drawn based on the three core research questions.

First, the validity of threshold-based EPE detection was confirmed. The results demonstrate that methodology based on the Generalized Pareto Distribution (GPD), specifically the AD and MT-AD approaches, is highly effective for detecting extremes in the complex terrain and climatic heterogeneity of Southern India. While the simple percentile technique tends to uniformly overestimate threshold parameters, the statistical approaches successfully identify true heavy-tail events consistent with local climatic characteristics, as discussed by Sree Anusha and Maheswaran (2025).

Second, a precipitation network for the southern peninsular region of India was constructed, revealing a source-to-sink architecture. The Western Ghats acts as a potent localized orographic source. However, it exhibits low connectivity due to topographical isolation. Conversely, the North Interior Karnataka (NIK) region is identified as a synoptic propagation hub with high divergence and degree values, serving as a primary corridor that transmits extreme precipitation signals to the southeastern inland sink regions.

Third, ENSO phases are identified as a critical factor modulating the network topology. During the El Niño phase, the propagation system becomes fragmented with a significant reduction in connectivity. In contrast, the La Niña phase facilitates the formation of organized and robust propagation pathways, a finding further supported by the 2022 case study. The total degree during La Niña is approximately 1.5 times higher than that during El Niño, highlighting a distinct shift in propagation dynamics according to the climate phase.

In summary, this study demonstrates that climate network metrics are useful for quantifying the propagation mechanisms of extreme precipitation in Southern India.

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